What Missionary Control Left in the Language of Two 1880s isiXhosa Newspapers

Emmanuel Azubuike (2698752) George Ayensu (2722188) Banele Mkhize (2663814)
University of the Witwatersrand, Johannesburg, South Africa

Abstract

From 1884 the missionary *Isigidimi sama-Xosa* and the African-owned *Imvo Zabantsundu* ran side by side. Did missionary control show in *what* they printed or *how* they used words? Imvo was not more political in vocabulary: of ten political concepts fixed in advance, only parliament differs (3.3×, uncorrected). Exploratory religion differs most: Isigidimi uses it four times as often, in readers' letters too, and the gap holds both in the same years and under the same editor, Jabavu. Framing evidence is weak and not specific to political words. Old spelling penalises only the African-language tokenizer. The results are consistent with control shaping what the papers printed; whether it shaped how they wrote lies below our detection floor.

Introduction and Background

Where did missionary control leave its trace in the language of the earliest isiXhosa press? In the IsiXhosa Intellectual Traditions archive (Schoots et al., 2023a), *Isigidimi sama-Xosa* was a missionary paper; *Imvo Zabantsundu*, founded in 1884 by Isigidimi's former editor J. T. Jabavu, was the first African-owned isiXhosa paper, free "from any missionary control" (Schoots et al., 2023a, p. 55). From late 1884 both had African editors, Gqoba and Jabavu (Schoots, 2023b, p. 71). Comparing the papers contrasts control, but also two editors, places and audiences, which one paper per side cannot separate. So we also hold the editor constant: Jabavu edited Isigidimi in 1881–84, then Imvo.

Communication research separates *agenda-setting*, what a paper covers (McCombs & Shaw, 1972), from *framing*, how it presents it (Entman, 1993). We measure the first as **salience**, how often a topic's words appear, and the second with word embeddings (Hamilton et al., 2016a). Such studies mostly use modern news (Card et al., 2015; Field et al., 2018); Schoots (2023b) counted root words in these papers. We add controls that make both testable, the same-editor contrast, and a letters-versus-rest split to ask *how* control worked; finally, we ask whether this history still shapes how machines read the archive (Petrov et al., 2023; Rust et al., 2021). **H1**: the papers differ in how often they use political and religious words. **H2**: Isigidimi's political words sit closer to religious contexts. **H3**: historical text and its spelling raise tokenizer cost.

Methods

Data. We use the archive's human-corrected transcriptions, not spell-checked (Schoots et al., 2023a, p. 59), from November 1884 to 1888: 286,663 tokens of Isigidimi and 1.31M of Imvo.

Salience. Concepts are word stems adapted from Schoots (2023b) (Table S1): ten political concepts fixed in advance, four neutral controls, and religion (*tixo*, *kristu*). Religion is exploratory: the archive's authors expect it (Schoots et al., 2023a, p. 57), but a classifier trained on the same years chose it (logistic regression on shared words, issue-grouped nested cross-validation, 0.83 balanced accuracy). Ratios carry issue-bootstrap CIs; Jabavu's Isigidimi runs from January 1881 to October 1884.

Threat	Response
Different years	Nov 1884–1888 only; month-matched draws
Imvo 4.6× larger	Ten Imvo samples of Isigidimi's size
Frequency effects	2000 matched random word sets (Monroe et al., 2008)
Change over time	Each paper's own drift in other years
Unstable embeddings	Tune for seed stability; average ten runs
Non-political effects	Four neutral control concepts
One paper per side	Same editor in both; place, audience unfixed

Table 1: Threats to a fair comparison, and our response.

Letters. A letter opens by addressing the editor (*nkosi mhleli … kaundifakele*, 'sir, editor … please print') and ends with a signature; 28 of 30 random detections were genuine letters (Table S5).

Framing. Embedding shift is a weak proxy for Entman's framing: it also reflects sense, grammar and topic. We train fastText embeddings (Bojanowski et al., 2017) tuned for stability across seeds, since unstable embeddings create false shifts in small corpora (Antoniak & Mimno, 2018); this favours a grammar-heavy window of 1, so we also report windows 2 and 5. A word's *profile shift* is $1 - r$ between its similarities to 500 common words in each paper (Hamilton et al., 2016b); *neighbour overlap* compares its ten nearest neighbours (Gonen et al., 2020); chance comes from shuffling issues (Dubossarsky et al., 2017). The directional test, aimed at framing, asks whether political words sit nearer religious or political anchors (Table S4); to read it, we count religious and legal words near *umhlaba*.

Tokenizers. We count subwords for XLM-R (Conneau et al., 2020), Serengeti (Adebara et al., 2023) and mBERT (Devlin et al., 2019) on modern news (Adelani et al., 2023), on the same news in old spelling (per word) and on our text (per character).

Results

H1: different things, but not more politics. Of ten political concepts, only parliament differs: Imvo uses it 3.3 times as often ($p = 0.018$ against matched sets; Fig. 1); law, vote, rights, land and government do not. Exploratory religion does: Isigidimi uses it 4.2 times as often ($p = 0.020$). Neither survives correction for 15 tests ($p < 0.003$). (CIs ask whether rates differ; matched sets, whether a concept differs more than random words.) The religious gap holds in letters (0.33, CI 0.17–0.65) as in the rest (0.22). Under the same editor, Jabavu, Imvo carried religious words a third as often as Isigidimi (0.34,

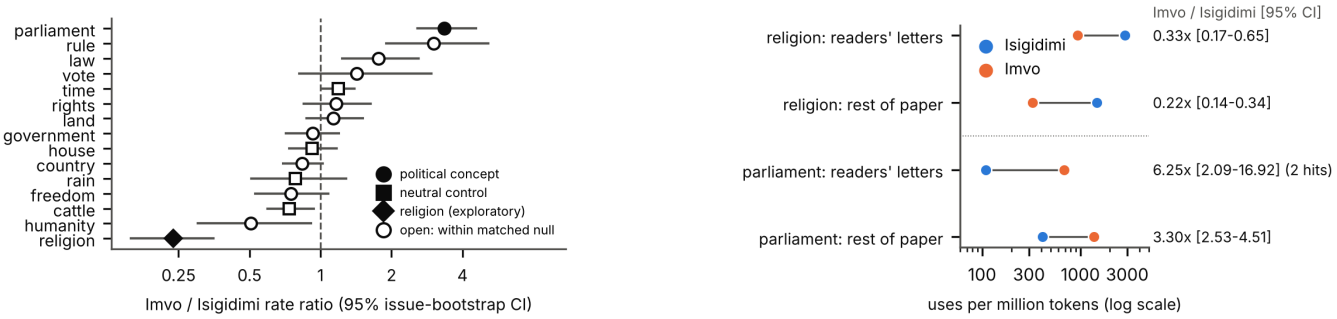


Figure 1: What the papers printed. Left: Imvo/Isigidimi rate ratio (95% CI; filled: $p < 0.05$ against matched sets, uncorrected). Right: the religious gap holds in readers' letters too.

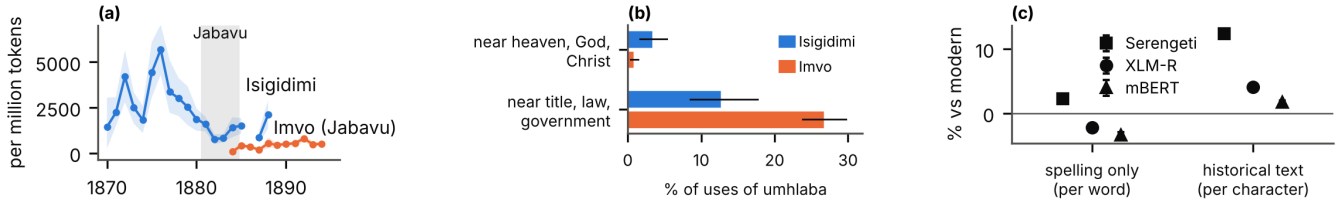


Figure 2: (a) Religious words per million tokens; grey: Jabavu edits Isigidimi, then Imvo; Gqoba edits Isigidimi from late 1884. (b) Uses of *umhlaba* near religious or legal words (95% CI). (c) Extra subwords against modern text (95% CIs smaller than markers).

CI 0.24–0.47, matched-set $p = 0.038$), and parliamentary words 1.7 times as often ($p = 0.096$). Under Gqoba, Isigidimi carried religious words 1.45 times as often as under Jabavu ($p = 0.148$). Within Isigidimi, religious words had already fallen to 0.31 of their 1876–80 level by 1881–84, while parliament rose 50-fold and vote 12-fold ($p \leq 0.007$; Fig. 2a; Table S7).

H2: framing differences are weak and not specific. Word use differs overall (profile shift 0.348 vs 0.230 by chance); no concept stands out: parliament reaches the 96th percentile of matched sets, the control *cattle* 90th–99th (Fig. S1). Political words lean towards religious anchors in Isigidimi in all five settings (Δ +0.44 to +0.76 z-units; Bonferroni-significant in three; Table S2). But without *land* the lean is rarely significant ($p = 0.010$–0.256); the control *house* leans further (+0.84 to +1.51); dropping the anchor *endlu* ('in the house') leaves both leans in place; and *umhlaba* sits near religious words in 3.4% of Isigidimi's uses but 0.9% of Imvo's (Fig. 2b). A one-word concept must shift by 0.15 (1.645 matched-set SDs) to be detected, more than the overall gap (0.118).

H3: historical text costs more. Per character, our text needs 12.4%, 4.1% and 1.7% more subwords with Serengeti, XLM-R and mBERT; old spelling alone raises only Serengeti's cost (2.2% per word; Fig. 2c).

Discussion

Is it control, or time? (H1). Religious words were falling in Isigidimi anyway, so we triangulate across a 2×2 of control and editor with one cell missing. Imvo against Gqoba's Isigidimi fixes the years (0.24); Jabavu's two papers fix the editor (0.34). Both show the gap against matched sets. Were the gap only the trend, Isigidimi would have kept falling after 1884; instead it held or rose (1.45). A Jabavu-era Imvo as religious as his Isigidimi, or a further fall under Gqoba, would have refuted control; neither occurred. Politics instead followed the times, rising within Isigidimi before Imvo existed. We cannot judge whether Jabavu was right to leave; leaving brought less religion, not clearly more politics. The gap reached beyond editors: Isigidimi's letter-writers used religious words three times as often as Imvo's, so the papers drew on, or chose, different writers. One wrote that *ilizwi alizanga ukuba silipate, lize ukuba lisipate* ('the Word did not come for us to carry it; it came to carry us', or 'rule us'; 1 Nov. 1884; our gloss, for speaker review).

What the embeddings see (H2). The framing signal looks like a difference in meaning, not slant: Isigidimi more often uses *umhlaba* for the earth, Imvo for land under law, which alone would move the word in each model. The detection floor makes the null a bound: large framing differences are ruled out, small ones are not.

Responsible NLP (H3). The papers' history also reaches today's tools: written isiXhosa took shape in the mission press (cf. Gilmour, 2007), and its old spelling penalises only the African-language model, perhaps because Serengeti's vocabulary has twice XLM-R's share of *th*/*kh*/*ph* pieces (1.9% vs 0.9%); a one-model hypothesis. A normalised working copy must not replace the source: the mapping is lossy (old *t* may be modern *t* or *th*). Our results describe two papers, not isiXhosa speakers; cultural readings without speaker review would be extractive (Bird, 2020).

Limitations and conclusion. Two papers, one window, a missing year (1886), an incomplete 2×2, a heuristic letter split, a news-domain mismatch, unreviewed glosses (e.g. *rulumente*; Schoots et al., 2023a, p. 62). Within these limits, the evidence fits control shaping what the papers printed, down to their letter-writers; any effect on political language lies below our detection floor.

References

Adebara, I., Elmadany, A., Abdul-Mageed, M., & Alcoba Inciarte, A. (2023). SERENGETI: Massively multilingual language models for Africa. In *Findings of the Association for Computational Linguistics: ACL 2023.*

Adelani, D. I., et al. (2023). MasakhaNEWS: News topic classification for African languages. In *Proceedings of IJCNLP-AACL 2023*.

Antoniak, M., & Mimno, D. (2018). Evaluating the stability of embedding-based word similarities. *Transactions of the Association for Computational Linguistics*, *6*, 107–119.

Bird, S. (2020). Decolonising speech and language technology. In *Proceedings of COLING.*

Bojanowski, P., Grave, E., Joulin, A., & Mikolov, T. (2017). Enriching word vectors with subword information. *Transactions of the Association for Computational Linguistics*, *5*, 135–146.

Card, D., Boydstun, A. E., Gross, J. H., Resnik, P., & Smith, N. A. (2015). The media frames corpus: Annotations of frames across issues. In *Proceedings of ACL-IJCNLP.*

Conneau, A., Khandelwal, K., Goyal, N., et al. (2020). Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics.*

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT* (pp. 4171–4186).

Dubossarsky, H., Weinshall, D., & Grossman, E. (2017). Outta control: Laws of semantic change and inherent biases in word representation models. In *Proceedings of EMNLP.*

Entman, R. M. (1993). Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, *43*(4), 51–58.

Field, A., Kliger, D., Wintner, S., Pan, J., Jurafsky, D., & Tsvetkov, Y. (2018). Framing and agenda-setting in Russian news: A computational analysis of intricate political strategies. In *Proceedings of EMNLP.*

Gilmour, R. (2007). Missionaries, colonialism and language in nineteenth-century South Africa. *History Compass*, *5*(6), 1761–1777.

Gonen, H., Jawahar, G., Seddah, D., & Goldberg, Y. (2020). Simple, interpretable and stable method for detecting words with usage change across corpora. In *Proceedings of ACL*.

Hamilton, W. L., Leskovec, J., & Jurafsky, D. (2016a). Diachronic word embeddings reveal statistical laws of semantic change. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics* (pp. 1489–1501).

Hamilton, W. L., Leskovec, J., & Jurafsky, D. (2016b). Cultural shift or linguistic drift? Comparing two computational measures of semantic change. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 2116–2121).

McCombs, M. E., & Shaw, D. L. (1972). The agenda-setting function of mass media. *Public Opinion Quarterly*, *36*(2), 176–187.

Monroe, B. L., Colaresi, M. P., & Quinn, K. M. (2008). Fightin' words: Lexical feature selection and evaluation for identifying the content of political conflict. *Political Analysis*, *16*(4), 372–403.

Petrov, A., La Malfa, E., Torr, P. H. S., & Bibi, A. (2023). Language model tokenizers introduce unfairness between languages. In *Advances in Neural Information Processing Systems.*

Rust, P., Pfeiffer, J., Vulić, I., Ruder, S., & Gurevych, I. (2021). How good is your tokenizer? On the monolingual performance of multilingual language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*.

Schoots, J. (2023b). Analyzing political formation through historical isiXhosa text analysis: Using frequency analysis to examine emerging African nationalism in South Africa. In *Proceedings of the Fourth Workshop on Resources for African Indigenous Languages (RAIL 2023)* (pp. 65–75).

Schoots, J., Ngwendu, A., De Wet, J., & Muftic, S. (2023a). IsiXhosa intellectual traditions digital archive: Digitizing isiXhosa texts from 1870-1914. In *Proceedings of the Fourth Workshop on Resources for African Indigenous Languages (RAIL 2023)* (pp. 54–64).

Statement on the use of generative AI (Faculty of Science AI Ethics Statement)

Generative AI (Claude, by Anthropic, used through the Claude Code tool) helped to write the analysis code (data preparation, embedding experiments, statistics, figure scripts), to draft and edit the text and LaTeX source, and to fill in the checklist below from the course template. Generative AI was also used to review drafts and suggest revisions; suggested changes were checked against the code outputs and the cited sources before being kept. All numbers in the paper are produced by the submitted code and were read from its output files. The authors take full responsibility for all code and text submitted, including the accuracy of every claim and citation.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state three hypotheses (H1 salience, H2 framing, H3 tokenizer cost) and report what was found: of ten pre-specified political concepts only parliament differs (uncorrected); the exploratory religious gap holds in readers' letters, in the same years and under the same editor (Jabavu), while the editor change within Isigidimi is not distinguishable from random words; political vocabulary followed the times; and the negative results (framing is weak and non-specific; the directional lean survives correction in three of five settings, but in at most one without land, and a neutral control leans further; the lean rests on umhlaba, which the two papers use in different senses). Claims about meaning are scoped as unreviewed by a first-language speaker.

   Guidelines:

   - The answer NA means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Discussion ends with limitations (two newspapers, one window, 1886 Isigidimi editions missing, an incomplete control-by-editor design, a heuristic letter split, a modern-news domain mismatch, unreviewed stems and glosses; the Results flag the one-model tokenizer hypothesis, and Table S3 the era-confounded held-out windows) and states the detection floor of the framing test.

   Guidelines:

   - The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [NA]

   Justification: The paper has no theoretical results.

   Guidelines:

   - The answer NA means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.

- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The Methods give the window, the date-matched subsampling, concept stems, model and hyper-parameter choices, the permutation null, the matched-set tests and the tokenizer conditions. Anchor lists are summarised in Table S4; word-joining and tokenizer rewrite rules are in the submitted code and README.txt, which gives the run order; seeds are fixed.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
   (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
   (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
   (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
   (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The code is submitted with requirements.txt and README.txt (exact run order). The newspaper texts are the public IsiXhosa Intellectual Traditions archive: the course-provided zip is read by src/01_prepare_data.py and the same texts can be downloaded from the archive DOIs listed in README.txt (Schoots et al., 2023a); the modern text is public (MasakhaNEWS) and README.txt gives the download command.

   Guidelines:

   - The answer NA means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://nips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so "No" is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://nips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental Setting/Details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Hyper-parameters were chosen on text outside the test window (Isigidimi 1876-83, Imvo 1889-94) by seed stability over four grid rounds (outputs/tuning_grid.csv), the classifier uses issue-grouped nested cross-validation, and the subsampling and permutation procedures are described.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment Statistical Significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Salience has issue-level bootstrap CIs; framing uses a permutation null (50 permutations) and 2000 frequency-matched word sets; tokenizer changes have bootstrap CIs (within about 0.5%, stated in the Results). The 15 salience tests are uncorrected and none survives Bonferroni ($p < 0.003$); the framing tests are reported uncorrected, except the directional test, which survives Bonferroni over five settings ($p < 0.01$) in three, but in at most one without land, and a neutral control leans further; the paper rests its argument on robustness and corroboration. The held-out comparison is era-confounded and is presented as an era-sensitivity check, with within-paper drift (including vote's large drift) as calibration; the classifier was trained on the test window, so religion is exploratory. The same-editor contrasts are tested against matched sets as well as bootstrap CIs; the letters/rest comparison and the senses of umhlaba use issue-bootstrap CIs only; the letter detector was checked by hand on 30 random detections (28 genuine), and its recall was not measured.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
- If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments Compute Resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments ran on CPU only (no GPU) in roughly three hours in total, including tuning rounds and the earlier framing design that was replaced; README.txt gives the four-core setup.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code Of Ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The work uses public-domain historical newspapers and public modern news text, collects no data from people, and flags that interpretation needs first-language-speaker review. To our knowledge it conforms to the Code of Ethics.

Guidelines:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader Impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: The Discussion covers fairness implications (higher tokenizer cost for historical isiXhosa, worst for the African-language model; the case against replacing the source with a normalised copy) and the risk of over-reading statistical shifts as cultural meaning.

    Guidelines:

    - The answer NA means that there is no societal impact of the work performed.
    - If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

    Answer: [NA]

    Justification: No models or datasets with misuse risk are released: the outputs are statistics, word lists and code, and the source texts are public-domain archive material.

    Guidelines:

    - The answer NA means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: Assets are credited and licences stated: IsiXIT texts (Schoots et al., 2023a; public domain per its authors); MasakhaNEWS (Adelani et al., 2023; AFL-3.0 on its dataset card); XLM-R (MIT) and mBERT (Apache-2.0) per their Hub cards; Serengeti states no licence on its model card and is used only to count tokens; gensim (LGPL) and scikit-learn (BSD) for modelling. Versions are in requirements.txt.

    Guidelines:

    - The answer NA means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.

- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New Assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: New assets are the analysis code and derived word lists (anchors, word-joining rules, the speaker-check sheet), documented in README.txt; no new text corpus is released.

    Guidelines:

    - The answer NA means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [NA]

    Justification: No crowdsourcing or human-subjects research was conducted. A first-language isiXhosa speaker is to be consulted as an expert adviser to check our interpretations; their judgments are not reported as data.

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [NA]

    Justification: No human-subjects research was conducted.

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

Supplementary material

S1. Concepts, stems and the word forms they match (1 Nov 1884–1888, both papers)

Stems are matched as substrings of surface forms; following Schoots (2023b) we inspected forms for false positives (our first *law* stem also matched *inteto*, speech, and was dropped) but did not hand-filter them, so run-together forms (e.g. *rulumenteningomsebenzi*) and missed variants remain. All stems, forms and glosses await review by a first-language isiXhosa speaker. Political concepts were fixed before testing; religion was suggested by the classifier and is exploratory.

Concept	Type	Stems	Most frequent matched forms (count; number of distinct forms)
parliament	political	palamente	palamente (1754), ipalamente (47), palamenteni (22), epalamente (14), bapalamente (1), nepalamente (1), yepalamente (1); 11 forms
vote	political	voti	ivoti (301), voti (217), evotini (136), abavoti (98), ngevoti (55), levoti (22), nevoti (22); 115 forms
government	political	rulumente	rulumente (1882), rulumenteni (7), urulumente (5), serulumente (1), rulumenteningomsebenzi (1), zikarulumente (1), rulumenteayiko (1); 8 forms
rule	political	mbuso	zombuso (169), umbuso (76), mbuso (60), yombuso (42), kombuso (37), lombuso (28), kulombuso (23); 43 forms
law	political	mteto	umteto (724), lomteto (475), mteto (406), ngomteto (167), ngumteto (94), komteto (76), nomteto (76); 82 forms
rights	political	lungelo	amalungelo (315), ilungelo (288), lungelo (113), nelungelo (54), namalungelo (50), lilungelo (39), elilungelo (37); 115 forms
country	political	lizwe	lizwe (646), ilizwe (292), kwelilizwe (201), elilizwe (101), elizwe (96), belilizwe (76), yelilizwe (52); 78 forms
land	political	mhlaba	umhlaba (836), mhlaba (208), lomhlaba (156), nomhlaba (105), komhlaba (99), somhlaba (95), yomhlaba (94); 135 forms
freedom	political	kululek	inkululeko (27), kukululekile (24), ukululekile (13), ngenkululeko (10), abakulekileyo (8), nkululeko (8), nenkululeko (7); 92 forms
humanity	political	buntu	ubuntu (45), yobuntu (14), nobuntu (8), sobuntu (8), buntu (7), lobuntu (7), zobuntu (6); 29 forms
time	neutral	xesha	ixesha (787), ngexesha (364), xesha (298), amaxesha (243), lixesha (191), ngamaxesha (149), zamaxesha (114); 207 forms
house	neutral	ndlu	indlu (719), izindlu (328), kwindlu (276), ndlu (234), yendlu (227), ngendlu (133), nendlu (130); 193 forms
rain	neutral	mvula	imvula (267), mvula (31), yemvula (23), ezimvula (17), lemvula (16), yimvula (14), zimvula (10); 45 forms
cattle	neutral	nkomo	inkomo (555), zenkomo (258), nkomo (193), yenkomo (93), nenkomo (75), ngenkomo (35), lenkomo (33); 79 forms
religion	exploratory	tixo, kristu	tixo (684), kristu (207), makristu (7), kristukazi (7), mkristu (3), amakristu (2), lobukristu (1); 16 forms

S2. Secondary framing tests across settings

Percentile of the observed shift among 2000 frequency-matched word sets, and one-sided p (uncorrected). P+V = *palamente* + *voti*; P+V+R adds *rulumente* (uncertain meaning). Directional Δ: Isigidimi minus Imvo, religious minus political similarity (in z-score units), political concepts, two-sided p against matched sets; the last columns give the control *house*. Three settings survive Bonferroni over five ($p < 0.01$): window 1, window 2, and window 2 with forms joined. Joining partly disjunctive forms (*uku ba*) leaves the global profile gap unchanged. Matched sets are re-drawn for each test; without *land* the pooled set is smaller, so lower power may contribute to the higher p.

Setting	P+V profile	P+V Jaccard	P+V+R profile	P+V+R Jaccard	Δ	Δ p	p w/o land	house Δ	house p
window 1 (tuned)	92 (p=0.083)	96 (p=0.039)	92 (p=0.077)	99 (p=0.007)	+0.68	<0.001	0.010	+1.08	<0.001
window 1, forms joined	99 (p=0.010)	95 (p=0.053)	100 (p=0.001)	100 (p=0.002)	+0.44	0.016	0.088	+1.15	<0.001
window 2	86 (p=0.139)	94 (p=0.061)	90 (p=0.096)	99 (p=0.010)	+0.76	<0.001	0.045	+1.34	<0.001
window 2, forms joined	91 (p=0.094)	98 (p=0.023)	99 (p=0.011)	100 (p=0.002)	+0.58	0.009	0.064	+1.51	<0.001
window 5	62 (p=0.382)	98 (p=0.016)	81 (p=0.189)	100 (p=0.002)	+0.66	0.015	0.256	+0.84	0.061

S3. Era drift: does a concept's rate move across eras within one paper?

Per-million-token rates and ratios (later/earlier, or Imvo/Isigidimi) with issue-bootstrap 95% CIs.

Concept	Comparison	First	Second	Ratio (95% CI)
parliament	Isigidimi: later / earlier era	214	394	1.84 (0.96–4.98)
parliament	Imvo: later / earlier era	1320	758	0.57 (0.50–0.66)
parliament	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	214	758	3.54 (1.98–9.58)
government	Isigidimi: later / earlier era	1667	1088	0.65 (0.50–0.85)
government	Imvo: later / earlier era	1210	452	0.37 (0.31–0.45)
government	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	1667	452	0.27 (0.22–0.34)
vote	Isigidimi: later / earlier era	72	614	8.55 (3.51–18.94)
vote	Imvo: later / earlier era	799	921	1.15 (0.88–1.51)
vote	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	72	921	12.82 (7.85–25.92)
religion	Isigidimi: later / earlier era	2466	1556	0.63 (0.46–0.84)
religion	Imvo: later / earlier era	362	547	1.51 (1.15–2.03)
religion	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	2466	547	0.22 (0.18–0.28)

S4. Anchors for the directional test

Anchors are the 15 top-PMI neighbours of each seed pair in held-out text (Isigidimi 1876-83 + Imvo 1889-94), frozen before testing. Religious (15 words; seeds *tixo*, *kristu*): *sikelela*, *sindisa*, *ukumkanikazi*, *inkosazana*, *buka*, *oyingcwele*, *lilizwi*, *msindisi*, *nelizwi*, *kwelizwi*, *wanga*, *lika*, *yesu*, *izono*, *yehova*. Political (15 words; seeds *palamente*, *voti*): *emqulwini*, *wonyulo*, *lonyulo*, *elugqatsweni*, *eziqingata*, *lokunyula*, *kwesase*, *lokuvota*, *endlu*, *kwamagama*, *ukumela*, *izayo*, *ngalomteto*, *nawupina*, *kwabase*. Seed words cannot themselves be tested (Fig. S1). Political anchors include *endlu* ('in the house'), which might have favoured the neutral control *house*; re-running without *endlu* leaves both the political lean and *house*'s lean in place (outputs/framing_matched_tests_noendlu.csv), and *ngalomteto*, which contains the *law* stem. Glosses are unverified: two religious anchors (*ukumkanikazi*, *inkosazana*) may reflect royal rather than religious register, and several anchors on both sides (*lika*, *buka*; *nawupina*, *kwabase*, *izayo*) may be grammatical rather than topical, consistent with narrow-window PMI capturing grammar. All await speaker review.

S5. Readers' letters versus the rest of each paper

Letters are detected by src/12_letters.py: a letter opens by addressing the editor (vocative *mhleli* or *mcokeli*, not *u/ku/ngu mhleli*, which refer to him) or by asking for insertion (*ndifakele*, *kaundifakele*), and ends at the signature (a run of very short lines). A hand check of 30 random detections found 28 genuine letters; the errors were usifake umkala (horse advert: 'put a bit on it'); no mhleli wepepa lama bhulu ('and the editor of the Boer paper'). Recall was not measured. Imvo is restricted to the months in which Isigidimi has text; CIs from 4000 issue resamples; rates are not tested against matched sets. Hits = concept uses.

Part	Concept	Isig. tokens	Isig. hits	Imvo tokens	Imvo hits	Imvo/Isig.	95% CI
letters	parliament	18,344	2	70,475	48	6.25	2.09–16.92
letters	religion	18,344	52	70,475	66	0.33	0.17–0.65
letters	vote	18,344	4	70,475	58	3.77	1.04–18.93
letters	government	18,344	13	70,475	115	2.30	1.14–7.18
rest of paper	parliament	268,319	111	936,998	1280	3.30	2.53–4.51
rest of paper	religion	268,319	394	936,998	306	0.22	0.14–0.34
rest of paper	vote	268,319	172	936,998	821	1.37	0.79–2.83
rest of paper	government	268,319	299	936,998	902	0.86	0.67–1.13

S6. Which sense of *umhlaba*?

For every use of the land stem (*mhlaba*) we look at the 10 tokens either side. Religious cues: *izulu*, *lizulu*, *nezulu*, and words containing *zulwini*, *tixo*, *kristu*, *yesu*, *msindisi*, *ngcwele* (bare *zulu* is excluded: *kwa zulu* is Zululand). Property/political cues: words containing *taitile*, *mteto*, *rulumente*, *palamente*, *mantyi*, *fama*, *voti*, *nyulo*, *rafu*, *lokishi*. Most uses have neither cue. Example lines for a speaker to check are in outputs/land_kwic.md.

Paper	Cue	Uses	Share	95% CI
Imvo	religious	1408	0.9%	0.4–1.4%
Imvo	property/political	1408	26.8%	23.8–29.7%
Isigidimi	religious	353	3.4%	1.7–5.3%
Isigidimi	property/political	353	12.7%	8.5–17.7%

S7. Same editor, different control; same control, different editor

Cells (editors after Schoots, 2023b, p. 71): Jabavu edits Isigidimi (Jan 1881–Oct 1884); Jabavu edits Imvo (Nov 1884–1888); Gqoba edits Isigidimi (Nov 1884–1888); earlier editors of Isigidimi (1876–80). Ratios of rates per million tokens; 95% CIs from 4000 issue resamples; p: two-sided, against 2000 matched sets of random words (src/15_editor_control.py). Every contrast is also a contrast in time; *rule* has only 9 uses in Jabavu's Isigidimi, so its large ratios are not interpreted.

Concept	Jabavu: Imvo / Isigidimi			Isigidimi: Gqoba / Jabavu			Isigidimi: 1881–84 / 1876–80		
	ratio	95% CI	p	ratio	95% CI	p	ratio	95% CI	p
parliament	1.69	1.23–2.54	0.096	0.50	0.34–0.80	0.041	49.86	14.59–403.58	<0.001
vote	3.07	1.98–5.10	0.012	2.36	1.04–4.55	0.025	11.63	4.42–104.17	0.007
government	0.85	0.65–1.13	0.525	0.77	0.56–1.04	0.260	0.88	0.65–1.18	0.599
rule	19.47	9.52–76.87	0.002	8.28	3.61–31.65	0.015	0.38	0.09–1.36	0.227
law	1.36	1.07–1.71	0.210	0.80	0.54–1.17	0.378	1.31	0.97–1.77	0.343
rights	0.93	0.76–1.16	0.801	0.82	0.59–1.12	0.482	2.91	2.09–4.10	0.006
country	1.08	0.92–1.28	0.780	1.13	0.91–1.42	0.620	0.74	0.59–0.92	0.249
land	1.10	0.91–1.34	0.710	0.95	0.70–1.29	0.834	0.94	0.76–1.16	0.801
freedom	0.68	0.51–0.91	0.444	0.95	0.64–1.39	0.903	1.52	1.01–2.37	0.402
humanity	0.97	0.58–1.88	0.961	1.88	0.97–3.87	0.213	0.97	0.44–2.03	0.966
religion	0.34	0.24–0.47	0.038	1.45	1.06–1.98	0.148	0.31	0.23–0.42	<0.001

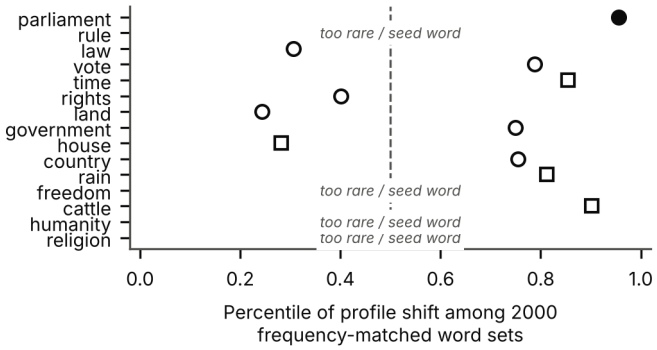


Figure S1: Profile-shift percentile of each concept among matched sets (markers as in Fig. 1, left); 'too rare' = too few occurrences, 'seed word' = anchor source, untestable.

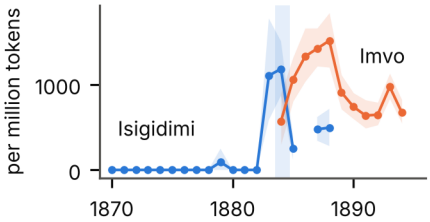


Figure S2: Parliamentary vocabulary per million tokens; it enters Isigidimi in 1883 (light blue: twice-monthly issues, Schoots et al., 2023a, p. 59).